7.5 Rotating in Three Dimensions

WORKSHEET

The question

STEP III 2022, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What does $\mathbf{n} \times \mathbf{r}$ look like geometrically, when $\mathbf{r}$ is perpendicular to $\mathbf{n}$? And when $\mathbf{r}$ is parallel to $\mathbf{n}$?

STAGE 1 Crossing twice

Ask yourself Write $\mathbf{r} = \mathbf{r}_\parallel + \mathbf{r}_\perp$, where $\mathbf{r}_\parallel = (\mathbf{n}.\mathbf{r})\mathbf{n}$ is the component along $\mathbf{n}$. What is $\mathrm{f}(\mathrm{f}(\mathbf{r}))$ in terms of these?

STAGE 2 The rotation

Ask yourself Compute $\mathrm{g}(\mathbf{n})$. Then compute $\mathrm{g}(\mathbf{r})$ for $\mathbf{r} \perp \mathbf{n}$, using $\mathrm{f}(\mathrm{f}(\mathbf{r})) = -\mathbf{r}$ for such $\mathbf{r}$.

STAGE 3 The reflection

Pause and try Simplify $\mathrm{h}(\mathbf{s}) = -\mathbf{s} - 2\mathrm{f}(\mathrm{f}(\mathbf{s}))$ using part (i). What does it do to vectors along $\mathbf{n}$, and to vectors perpendicular to $\mathbf{n}$?

Try this next

1. **Rotations in the plane.** (From STEP III, 2017.) Show that rotation by θ about the point a in the complex plane sends z to $e^{\mathrm{i}\theta}z + a(1 - e^{\mathrm{i}\theta})$. Show that rotating by θ about a and then by ϕ about b is a rotation by $\theta + \phi$ about some point c, provided $\theta + \phi$ is not a multiple of 2π. What is it when $\theta + \phi = 2\pi$?

2. **A specific rotation.** Find the matrix of the rotation by 120° about the axis in the direction $(1, 1, 1)$. Show that it permutes the coordinate axes cyclically.

3. **Powers of $\mathbf{K}$.** Show that $\mathbf{K}^2\mathbf{r} = (\mathbf{n}.\mathbf{r})\mathbf{n} - \mathbf{r}$ and $\mathbf{K}^3 = -\mathbf{K}$, and deduce Rodrigues' formula from the exponential series.

4. **Two reflections make a rotation.** Show that reflecting in two planes through the origin, at angle α to each other, is a rotation by 2α about their line of intersection.

Further reading

- **Simon Altmann, *Rotations, Quaternions, and Double Groups*** (Dover). The definitive account of three-dimensional rotations, including Rodrigues' contribution.

- **Simon Altmann, "Hamilton, Rodrigues, and the quaternion scandal"**, *Mathematics Magazine* 62 (1989). A short, lively history.
- **John Stillwell, *Naive Lie Theory*** (Springer). Rotations, quaternions and Lie groups for undergraduates.

The full guided solution, and what the question leads to at university, is in Section 7.5:
https://beyondstep.keithmcnulty.org/ch07/rotations-3d.html